MSA

October University for Modern Science & Arts
Faculty of Engineering
Department of Mechatronics System Engineering

Centralized grid-connected solar PV

Renewable Energy Project

Prepared By

Khaled Diaa Eldin Mohamed	247181
Abdelrahman Waeel Abdelmonem	244825
Ahmed Islam Ahmed	241649
George Girges Robile	254762
Mohamed Ahmed Mahmoud	248267
Sandra Naser Kerolos	244383
Beshoy Nabil Samir	240243

Supervised By

Associate Professor Dr. Mohammed Ali

Fall 2025

1. Aim of the Project

This paper attempts to investigate the thermodynamic properties of a photovoltaic solar panel and the extent of how operational temperatures affect the efficiency of energy conversion. The analysis aims at studying the generation of heat and the temperature distribution in a laminated photovoltaic module subjected to standard solar radiation, in an attempt to establish design-related variables that affect the efficiency.

2. Objectives of the Project

The Objective are as follows:

1. **Analyze conduction, convection, and radiation heat transfer.**
    - To investigate the thermal properties of a photovoltaic solar cell under the effect of solar irradiation.
2. **Compute temperature distribution and heat flux**.
    - Analyze how the operating temperature for solar cells is related to the conversion efficiency of photovoltaic cells.

3. Develop a 3D thermal model of a solar PV module.

- In order to simulate the temperatures throughout the multi-layer structure of a PV module.
- To explore possible design parameters that could help in reducing temperature increase and improve solar photovoltaic cell effectiveness.

4. Assign appropriate material properties for each layer.

- To explore how material properties and thicknesses of layers can affect heat transfer in the panel.

5. Validate simulation results using manufacturer reference data using ANSYS steady state thermal analysis.

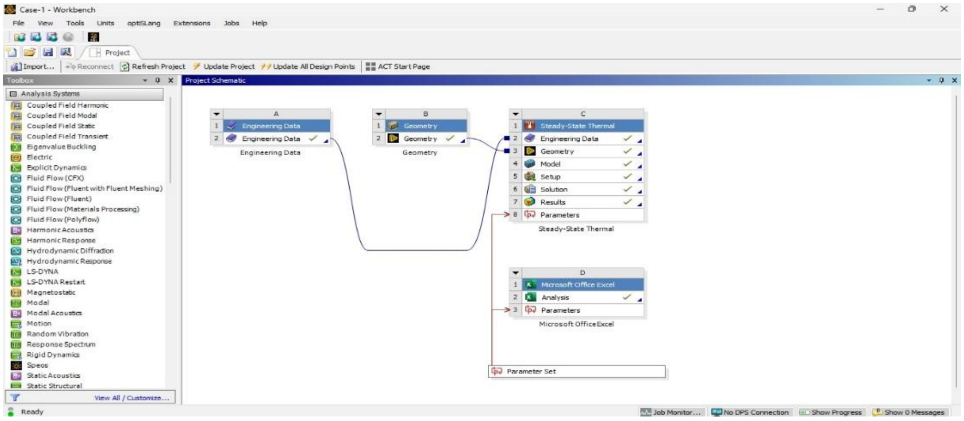


Figure 1: *Ansys workbench for the overall project*

3. Literature Review

Earlier reports indicate that the temperature of PV modules significantly affects electrical performance. FEM and CFD models have been developed to predict temperature rise and heat removal methods. References show that conduction is the dominant heat transfer inside the module layers, while convection and radiation drive the heat losses from those to environment. This software is often used because of its accuracy in coupled thermal problems and an ability to specify mixed convection and radiation boundary conditions.

4. Introduction

The increasing need for clean and sustainable energy sources has triggered a lot of innovation in renewable energy technologies, solar photovoltaic systems being at the forefront. These PV cells convert solar energy directly into electrical energy through semiconductor materials, most commonly silicon solar cells. Although a lot of innovation has taken place in solar PV technology, its efficiency still largely relies on working conditions, especially on temperature.

During the operation, only part of the solar energy is utilized to generate power, with the remaining part being lost in the form of heat. This causes an increase in the operating temperatures of solar cells. For silicon PV cells, even small increments in temperatures generate significant power losses. Therefore, thermal analysis plays an important role in improving the performance of solar cells.

Monocrystalline silicon solar cells are preferred due to their efficiency and reliable performance. These solar cells have several layers for protection, such as the front sheet, encapsulant layers, and the back sheet. Although the layers increase the robustness and stability of the solar cell, they also affect the thermal properties of the PV cell. The design, thickness, and type of layers have several critical effects on the thermal properties of the PV cell.

This research focuses on the thermodynamic aspects of a laminated photovoltaic solar panel. In this case, the research will evaluate the heat generation, conduction, and dissipation in the photovoltaic solar panel. With an analysis of temperature distribution in the solar panel, this research will address how design factors influence temperature and photovoltaic efficiency in a solar panel. This will mostly apply to rooftop power systems and solar power technology that receives high sunlight intensities over a relatively long period.

5.Energy Balance of the PV Module

The thermal behavior of the solar PV module is controlled by an energy balance between absorbed solar energy, thermal losses, and generated electrical power. The energy balance equation is expressed as:

Q (generated) = Q(solar)− (Q (th, loss, top) + Q (th, loss, bottom) + Q (op, loss))

Where:

- **Q (solar) is the incident solar irradiation.**
- **Q (th, loss, top) is the thermal loss from the top surface.**
- **Q (th, loss, bottom) is the thermal loss from the bottom surface.**
- **Q (op, loss) represents electrical losses.**
- **Q(generated) is the useful generated energy.**

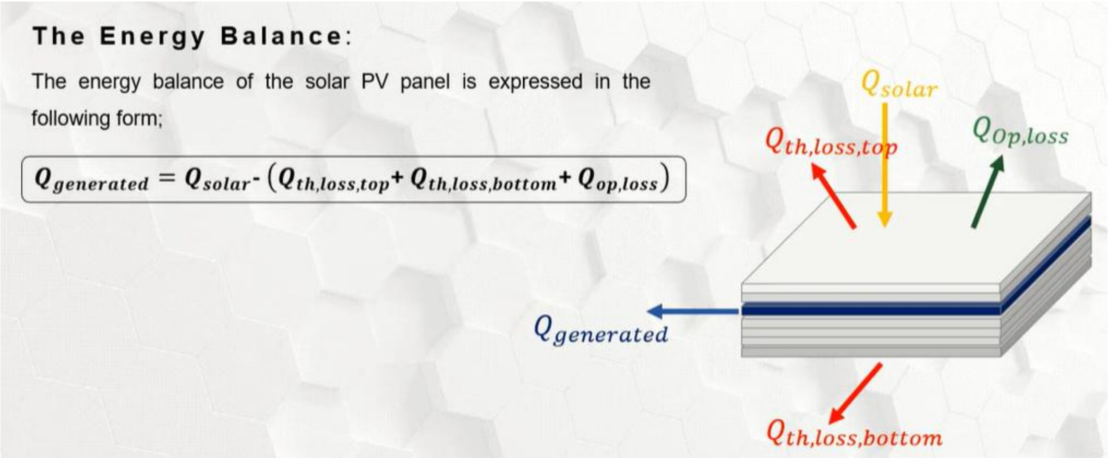


Figure2. *Energy balance of a solar PV module showing solar input, thermal losses from top and bottom surfaces, and generated electrical power.*

6. Geometry of the Photovoltaic Solar Panel

The PV module examined in this research is simulated as a flat, rectangular laminated structure consisting of various functional layers. It measures 1.153 m long, 0.518 m wide, and is a common format for a roof-top PV module. The active area is composed of 36 monocrystalline solar cells measuring 123 mm × 123 mm. Such cells are arranged evenly over the surface of the panel to ensure uniform exposure to sunlight and thermal cycles.

The module is built up in a seven-layer laminate with layers oriented perpendicular to the panel. Starting at the top, the layers are:

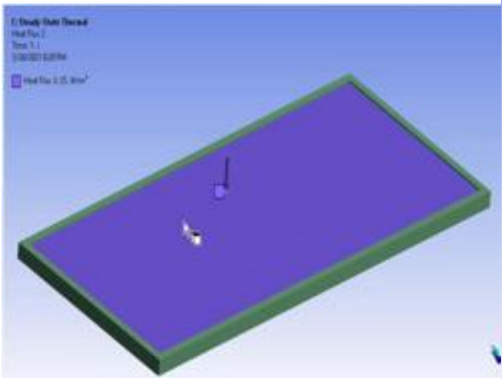
1) Front sheet thickness: 0.28 mm, made of ETFE which has optical transparency and environmental protection properties.

2) Thickness of 0.20 mm of EVA encapsulant, ensuring mechanical bonding and electrical insulation.

3) A 0.15mm thick monocrystalline silicon solar cell acting as the main layer for the photovoltaic conversion and heating process.

4) A secondary layer of encapsulant with a thickness of 0.20 mm made of EVA.

5) Thickness of 0.20 mm PET back sheet protecting the module from the environment from the rear side. 6) Adhesive tape layer, with a thickness of 0.13 mm, which adheres the laminated structure onto the support. 7) A support layer with a thickness of 2.00 mm made out of Carbon Fiber Reinforced Polymer (CFRP) to offer structural rigidity. In the analysis of the thermal phenomenon, the layers are treated as if they are homogenous and fully in contact, which helps to ensure that the heat flow is continuous. The heat flow is assumed to mainly occur in the thickness direction, with heat loss taking the form of convection and radiation on the front and back sides. This geometry is realistic for simulating the effect of the thermal phenomenon on the performance characteristics of the solar cell module.



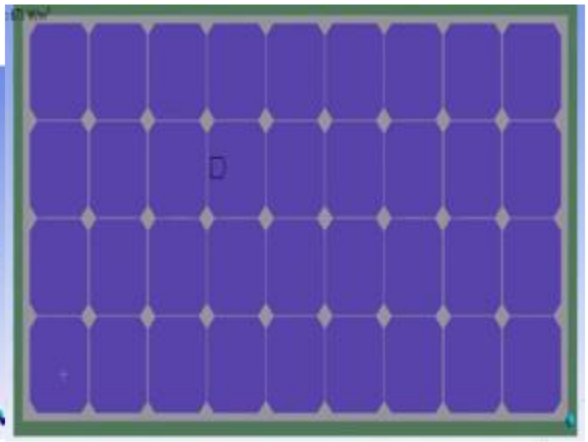


Figure 3: *geometry of the solar panel* ***Figure 4: the Pv cells Geometry***

7.Layers

Structural Composition and Layer Hierarchy

The solar panel assembly has been designed as a seven-layer stacked technology. This design has been focused on optimizing electrical efficiency, together with mechanical flexibility. Unlike the traditional solar panels designed as a rigid module, the current design utilizes a high-performance fluoropolymer front sheet designed for integration on curved substrates such as a car roof.

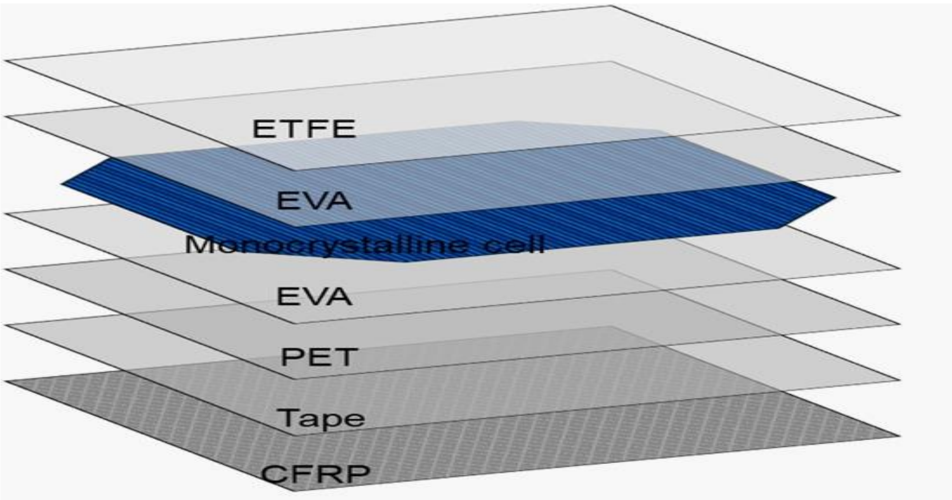
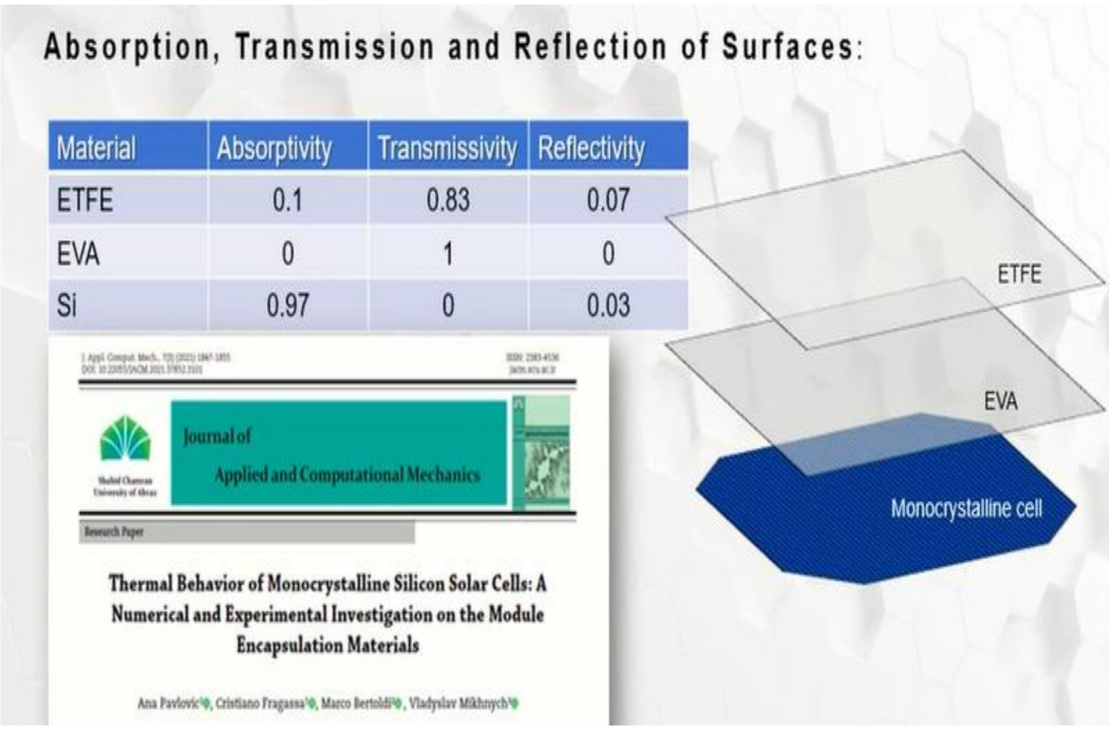


Figure 5: *the 7 layers details*



Material	Absorptivity	Transmissivity	Reflectivity
ETFE	0.1	0.83	0.07
EVA	0	1	0
Si	0.97	0	0.03

Figure 6: *the Absorption , Transmission and Reflection of surfaces*

so on.

Geometry models were build using the Design Modeler Module of Ansys by stacking 7 layers of 125x125mm, each one with proper material and thickness (Tab. 2 and 3). Specifically, five different models, one per layout, were developed. Layers were assembled considering *bonded contacts* which allow the complete transmission of forces and heat. The corners have been finally rounded off as in the real samples.

Discretization was implemented by SOLID186 by finite elements (FE). It is a higher-order solid element, defined by 20 nodes with 3 degrees of freedom (DoF) each (i.e., nodal translations x, y, z directions), that exhibits quadratic displacement behaviour. The system discretization was performed by (up to) 198060 nodes and 27702 elements.

Conditions and hypothesis used for the numerical thermal analysis can be summarized as follows:

- Whole radiance flux, equal to 1000 W/m^2, impacts the panel uniformly and orthogonally.
- First layer, in ETFE reflects 7% (= 70 W/m^2) and absorbs 10% (= 100 W/m^2) of the transmitted irradiance.
- Remaining energy (= 830 W/m^2) is applied as a constant heat flux on silicon cells (i.e., no application on EVA).
- Silicon layer reflects 3% toward ETFE (= 25 W/m^2) and absorbs all the rest (= 805 W/m^2).
- EVA is simplified as completely transparent matter.
- Heat is transferred between the layers by conduction.
- Heat transfer outwards occurs through ETFE and CFRP layers considering irradiation and convection phenomena.
- When ETFE is absent (3rd & 4th layouts), reflection and transmission are performed by the silicon layer.
- Emissivity, typically equal to the absorption coefficient (i.e., *Kirchhoff's Law*), is chosen as 0.89 for ETFE and CFRP.
- Convective heat transfer coefficient with respect to the ambient temperature is 5.67 W/m^2°C (eq. no wind conditions).
- Material properties are here considered independent from temperature.

3.3. Experiments

During the test, different panels were heated by appropriate equipment in controlled conditions of irradiation and environment, with temperatures measured over the entire surface. Specifically, thermal measures were done with respect to the 5 different layouts (Table 1) in 3 samples each. Tests were performed setting these 15 laminates on cardboard supports under solar lamps (fig.

Figure 7: *Emissivity for ETFE and CFRP*

8.Material Properties Table

The thermal and physical properties of the individual layers are critical for the simulation of the lamination process and the subsequent thermal performance of the panel

Materials	Density [kg/m3]	Thermal conductivity [W/m.K]	Specific heat [J/kg.K]
ETFE	1730	0.24	1172
EVA	945	0.35	2090
Silicon	2330	148	700
PET	1350	0.275	1275
CFRP	1490	6.83	1130
Tape	1012	0.19	2000

Figure8: *the engineering data of the layers*

9. Mesh

Meshing A mesh model was created with ANSYS Meshing. A fine mesh was employed near material interfaces to capture the temperature gradients correctly. The quality of the mesh (skewness and element size) was checked to provide numerical stability and accuracy.

The material selected in our work is the glass mixed with other materials, a cell of it made from silicon and support layer was made by aluminum.

All this stuff we didn't find in ANSYS, so based on engineering data, as a new material make a new density heat coefficient and Conductivity for every layer that we had we took from reference and then geometry and select every layer to create his material.

We chose quad element because we want our cell with a squares shape to that when the ANSYS use equations on cells that will be square and so everything used here became easy to do, also it takes the same form as I selected and the equation are right.

Mesh quality : in order to gets the highest right percent 0.997" =1%, there are the highest orthogonal quality that reach us.

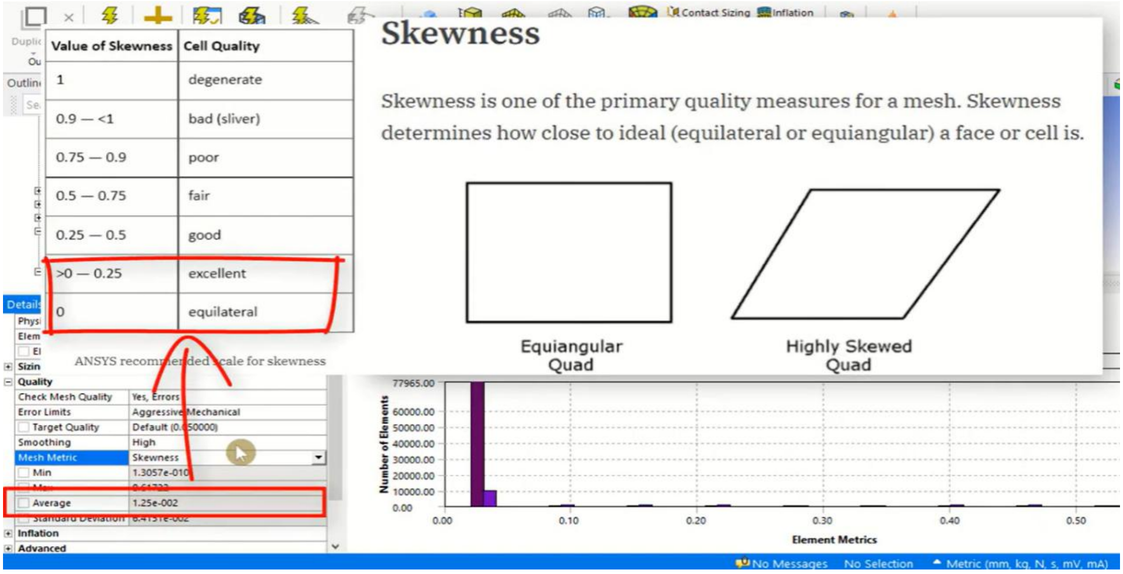


Figure 9: *Skewness of the Mesh*

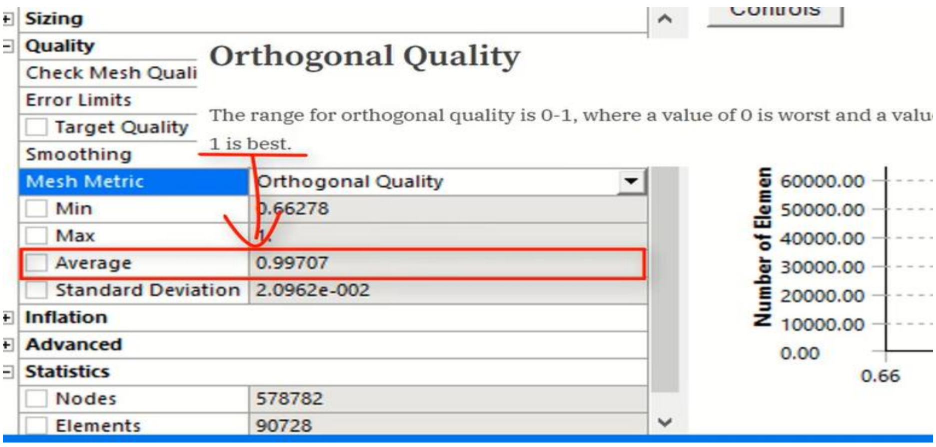


Figure 10: *Orthogonal Quality of the mesh*

.

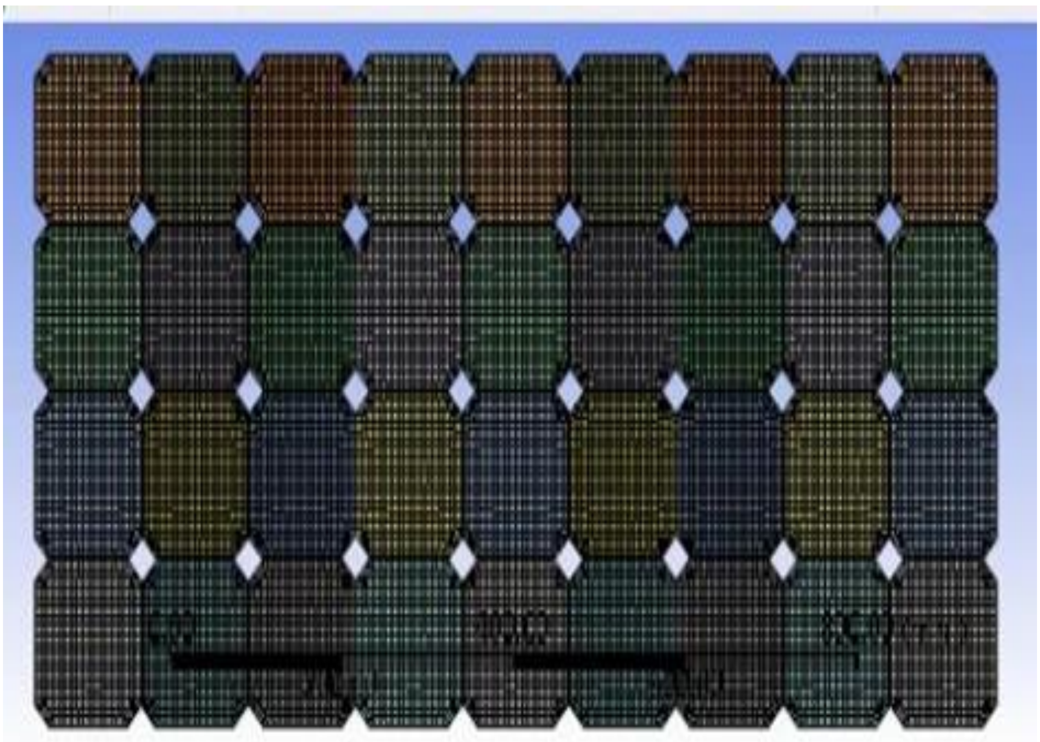


Figure11: *Finite element mesh of the solar PV*

10. Bonded Contacts

The reasons why we used bonded contacts between layers instead of other options like (shared topology, node merges, joints) are to make the layers glued together that, the pv panel wont slide or separate from each other's Also, other reasons to choose bonded is its simplicity. With bonded, we can keep a nice, clean mesh on each layer and let the software handle the connection. It gives us a much more stable model It really comes down to getting the most accurate stress data while keeping the simulation stable enough to actually get results. also, It keeps the math linear, which means we can run multiple design iterations like changing the layers thickness without making a complicated mathematical equations , makes it efficient and reducing simulation time. Furthermore, using bonded contact is really essential for capturing the thermal expansion behavior accurately. Since each layer, from the glass to the backsheet has its own different expansion coefficient.

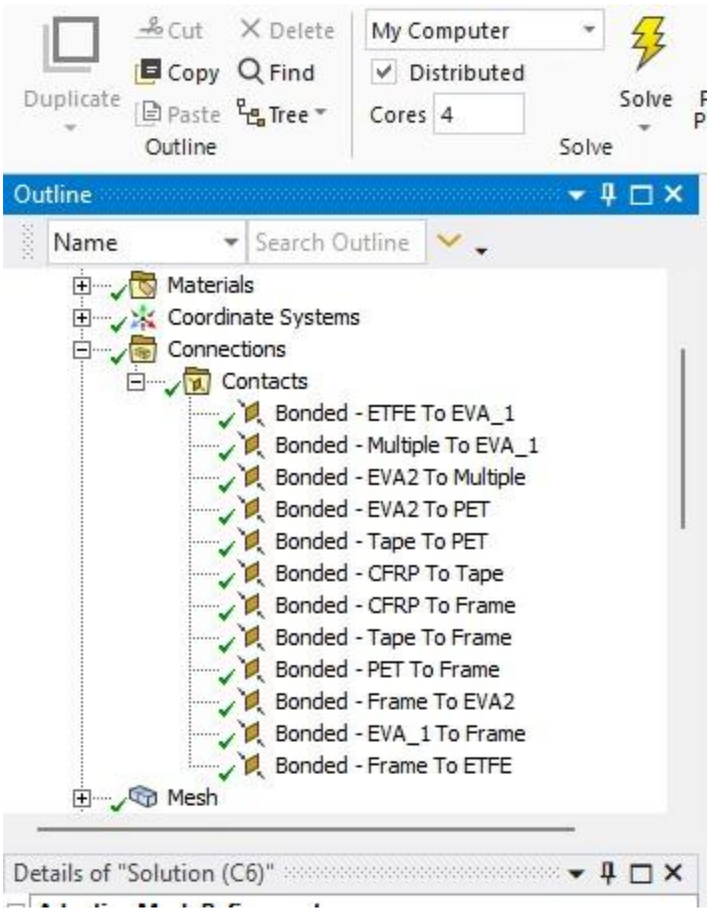


Figure12: *the contacts of bonds*

11. Heat Transfer Mechanisms

A) Conduction

Internal heat transfer in layers of materials (glass, EVA, silicon) according to Fourier's law. Thermal conductivities are given in ANSYS Engineering Data.

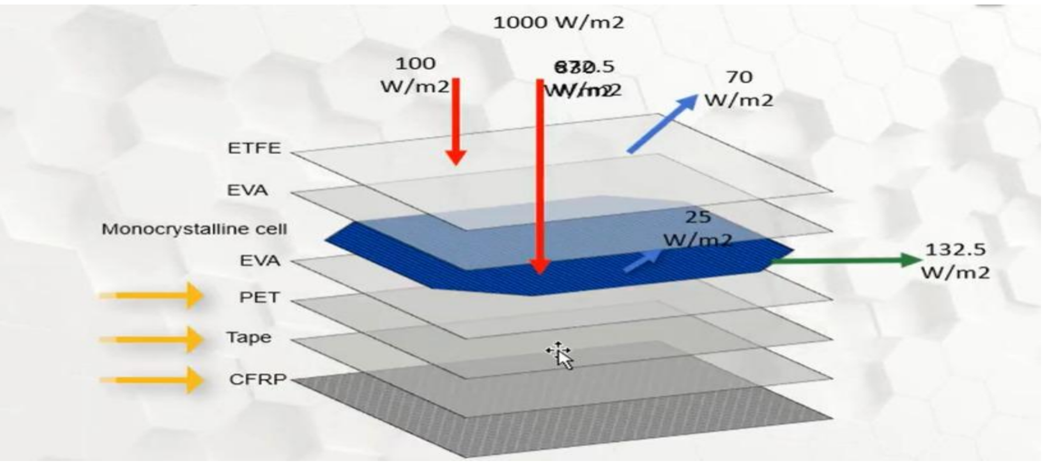


Figure 13: *Heat Transfer Mechanism*

B) Radiation

Environmental radiation to the surface using values for emissivity of outer surfaces and Stefan–Boltzmann law. This creates energy transfer between the panel surfaces and its surroundings.

C) Convection

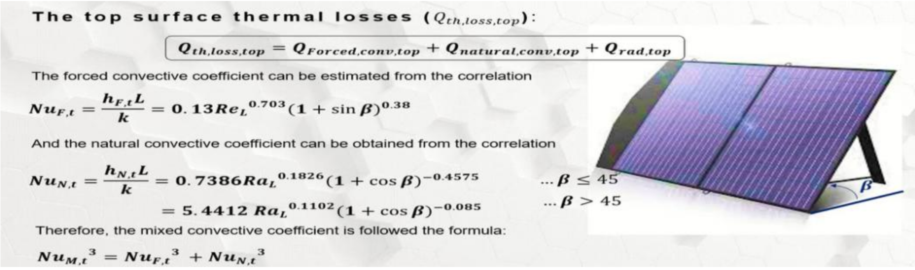
Heat transfer from the module surfaces to the ambient air due to convection is described by convection coefficients, typically obtained using empirical correlations for natural/forced convection.

Conduction, Convection and Radiation Thermal losses from the surfaces of PV modules are due to mixed convection (natural and forced).

- **Top Surface Thermal Losses**

Q (th, loss, top) = (Q (forced, conv, top) +Q (natural, conv, top) +Q (rad, top))

The forced convective heat transfer coefficient is estimated using:


The top surface thermal losses ($Q_{th,loss,top}$):

$$\boxed{Q_{th,loss,top} = Q_{Forced,conv,top} + Q_{natural,conv,top} + Q_{rad,top}}$$

The forced convective coefficient can be estimated from the correlation

$$Nu_{F,t} = \frac{h_{F,t}L}{k} = 0.13Re_L^{0.703}(1 + \sin\beta)^{0.38}$$

And the natural convective coefficient can be obtained from the correlation

$$Nu_{N,t} = \frac{h_{N,t}L}{k} = 0.7386Ra_L^{0.1826}(1 + \cos\beta)^{-0.4575} \quad \ldots\beta \leq 45$$

$$= 5.4412\ Ra_L^{0.1102}(1 + \cos\beta)^{-0.085} \quad \ldots\beta > 45$$

Therefore, the mixed convective coefficient is followed the formula:

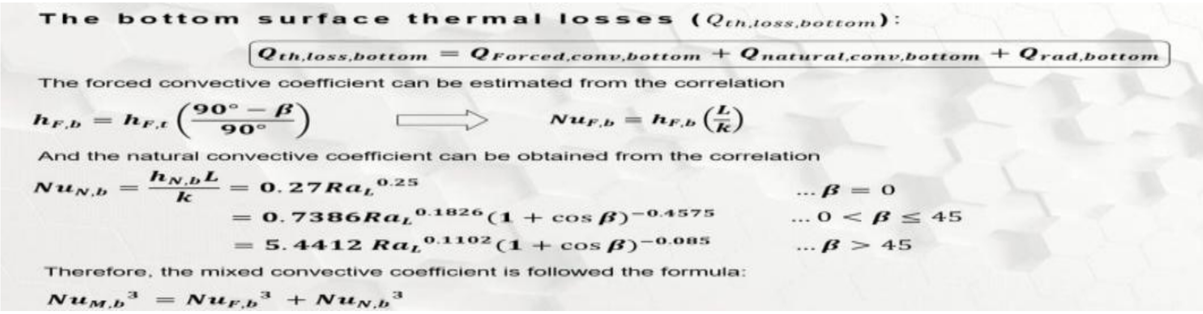
$$Nu_{M,t}^3 = Nu_{F,t}^3 + Nu_{N,t}^3$$

Figure 14: Top surface thermal loss mechanisms and convection correlations for a tilted PV module.

- **Bottom Surface Thermal Losses**

Q (th, loss, bottom) = Q (forced, conv, bottom) +Q (natural, conv, bottom) +Q (rad, bottom))

The forced convection coefficient for the bottom surface is adjusted based on tilt angle:


The bottom surface thermal losses ($Q_{th,loss,bottom}$):

$$\boxed{Q_{th,loss,bottom} = Q_{Forced,conv,bottom} + Q_{natural,conv,bottom} + Q_{rad,bottom}}$$

The forced convective coefficient can be estimated from the correlation

$$h_{F,b} = h_{F,t}\left(\frac{90^\circ - \beta}{90^\circ}\right) \Longrightarrow Nu_{F,b} = h_{F,b}\left(\frac{L}{k}\right)$$

And the natural convective coefficient can be obtained from the correlation

$$Nu_{N,b} = \frac{h_{N,b}L}{k} = 0.27Ra_L^{0.25} \quad \ldots\beta = 0$$

$$= 0.7386Ra_L^{0.1826}(1 + \cos\beta)^{-0.4575} \quad \ldots 0 < \beta \leq 45$$

$$= 5.4412\ Ra_L^{0.1102}(1 + \cos\beta)^{-0.085} \quad \ldots\beta > 45$$

Therefore, the mixed convective coefficient is followed the formula:

$$Nu_{M,b}^3 = Nu_{F,b}^3 + Nu_{N,b}^3$$

Figure 15: Bottom surface thermal loss mechanisms and convection correlations.

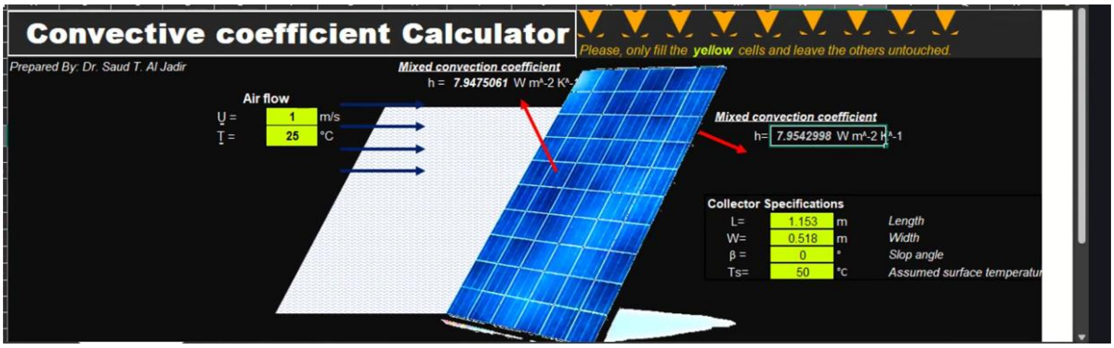


Figure 16: convective coeffcient calculator.

12. Solar Heat Flux

Also, we had to consider the **Heat Flux** in the simulation because it's a huge factor for PV performance. Basically, the heat flux represents the energy flow from the sun hitting the panel surfaces. We applied this to see how the thermal energy moves from the front glass, through the eva, and down to the backsheet. It is really important to track this because if the heat flux isn't distributed well, get hot spots that can actually degrade the solar cells much faster. Also, since we know that silicon cells lose efficiency when they get too hot, calculating the directional heat flux helps us understand if the panel design is good at heat gain or not. Calculations of heat flux at pv 7 layers changes depends on different factors like material and the thickness of each layer . the total heat flux on the pv was 1000 w/m^2. However, the full benefit was 132.5 w/m^2.

- heat flux at 100 W /m^2 (ETFE)
- heat flux at EVA = 830*0.03= 24.9
- heat flux at PV cells = 673W/m^2

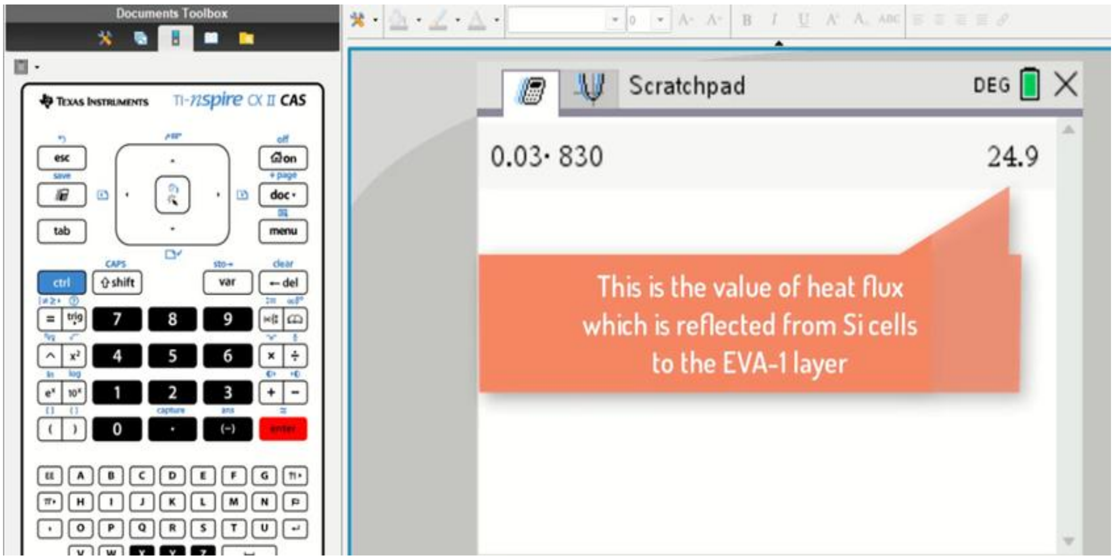


Figure 17: *Ansys Calculator for the heat flux*

13.Results

In the present work, we have focused on thermal process of the solar panel and analyzed that temperature is not uniform over the panel. From ansys results we see the edges of panel are 44°C but centre is higher (52–53°C).So, as we move slowly from edge to middle the temperature increment is very slow and middle portion is hottest in the parts. For the different layers of ' PV cells which is in touch with ETFE layer has a temperature from 46°C to 53°C and the CFRP back layer as well for That PV cells are more stressed to the heat than other layers.

The heat flux indicates heat transfers from -400 W/m² to -90 W/m² in Y direction. This is because heat moves from PV cells to lower layers and then the air that surrounds them. We can easily observe that heat flows primarily top to bottom, and then environment.

All the simulation were in real conditions that we used. The temperature of the environment is 25 °C The emissivity of the surface (with reference) \(\varepsilon = 0.89.\) The solar heat flux is imposed on the PV cells and the convection coefficient is about 7.95 W/m²·K. It can be similar to actual outdoor condition by the above conditions

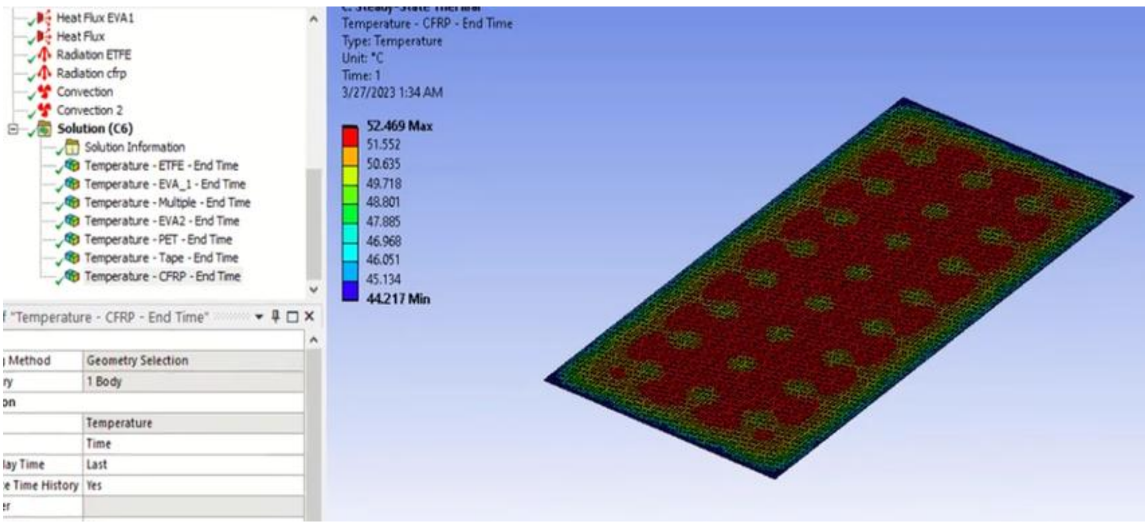


Figure 18: *Temperature distribution contour of the solar PV module CFRP*

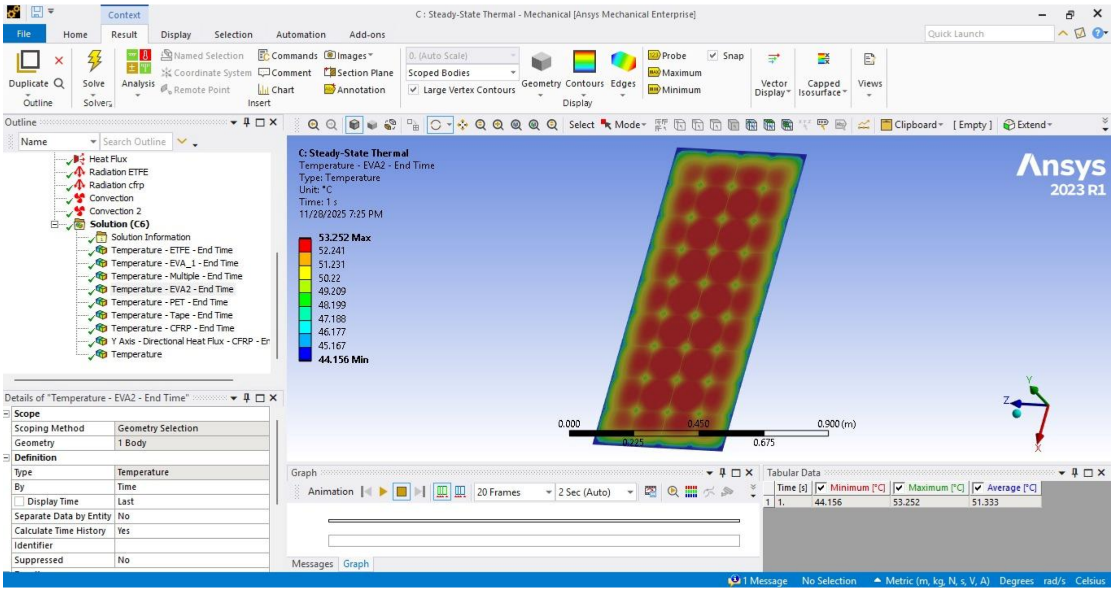


Figure 19: *Temperature distribution contour of the solar PV module EVA2*

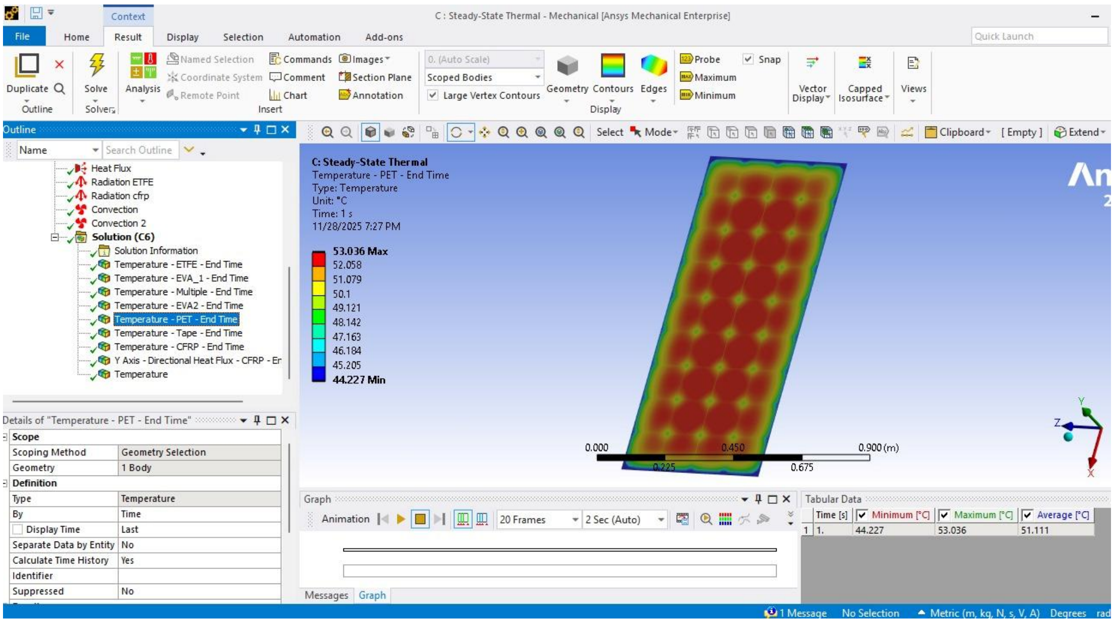


Figure 20: *Temperature distribution contour of the solar PV module PET*

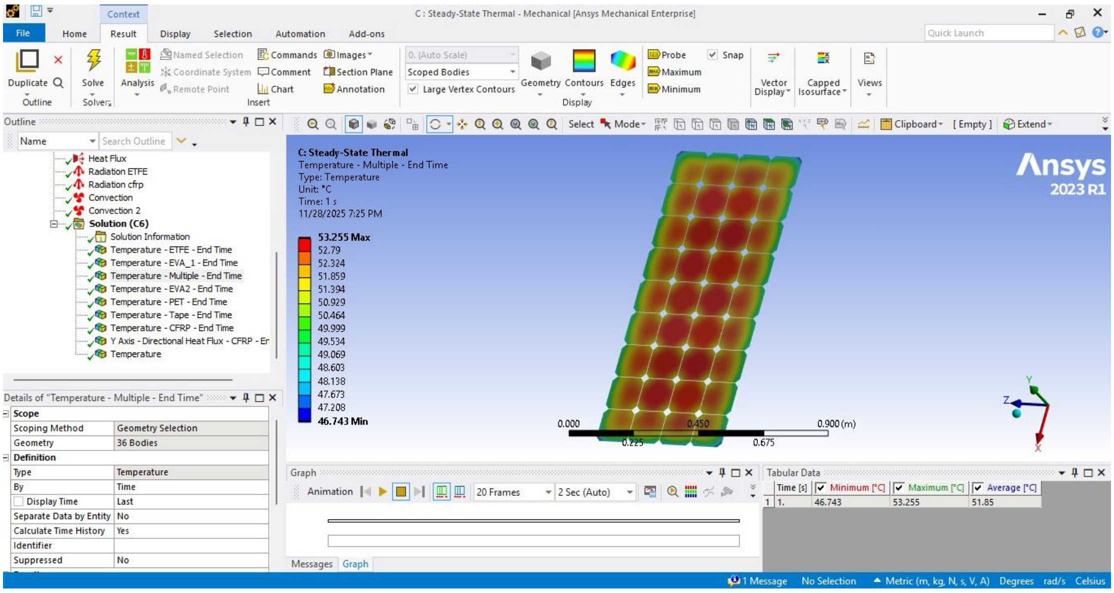


Figure 21: Temperature distribution contour of the solar PV module PV cells

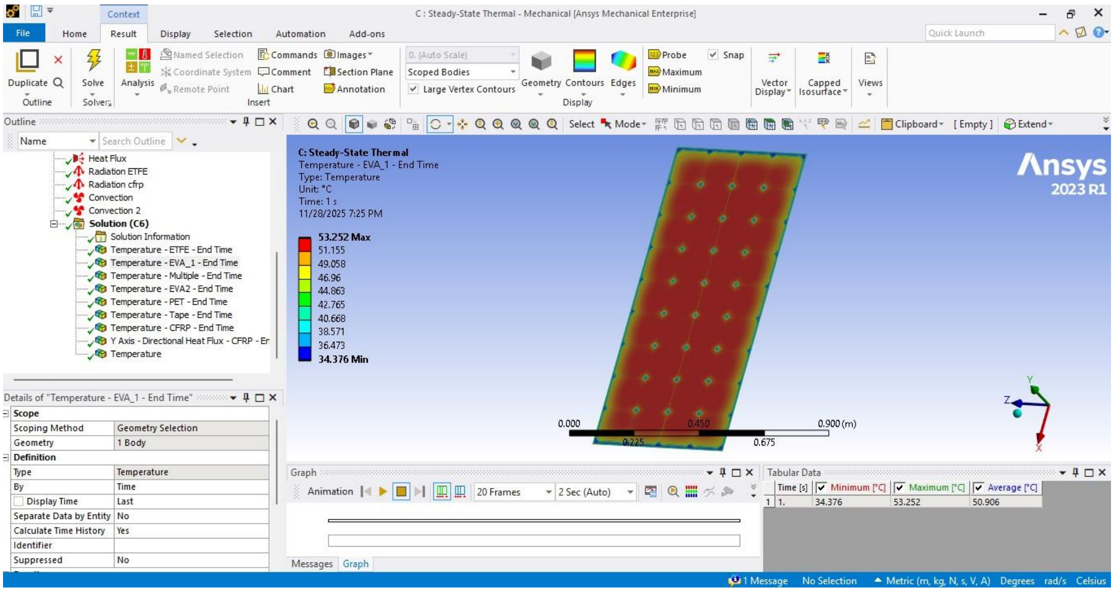


Figure 22: Temperature distribution contour of the solar PV module EVA1

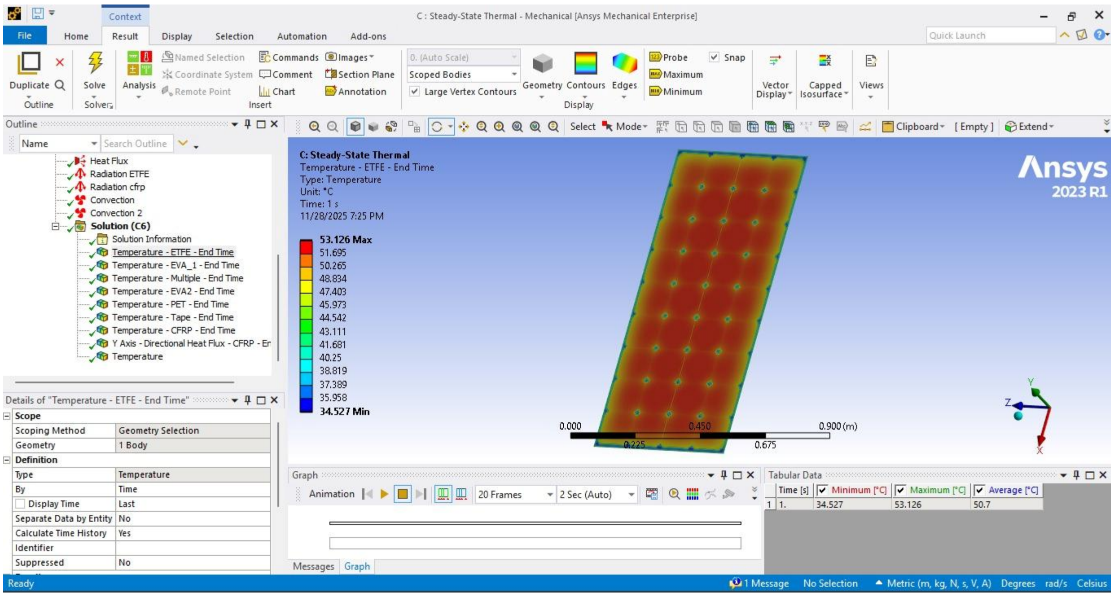


Figure 23: *Temperature distribution contour of the solar PV module ETFE*

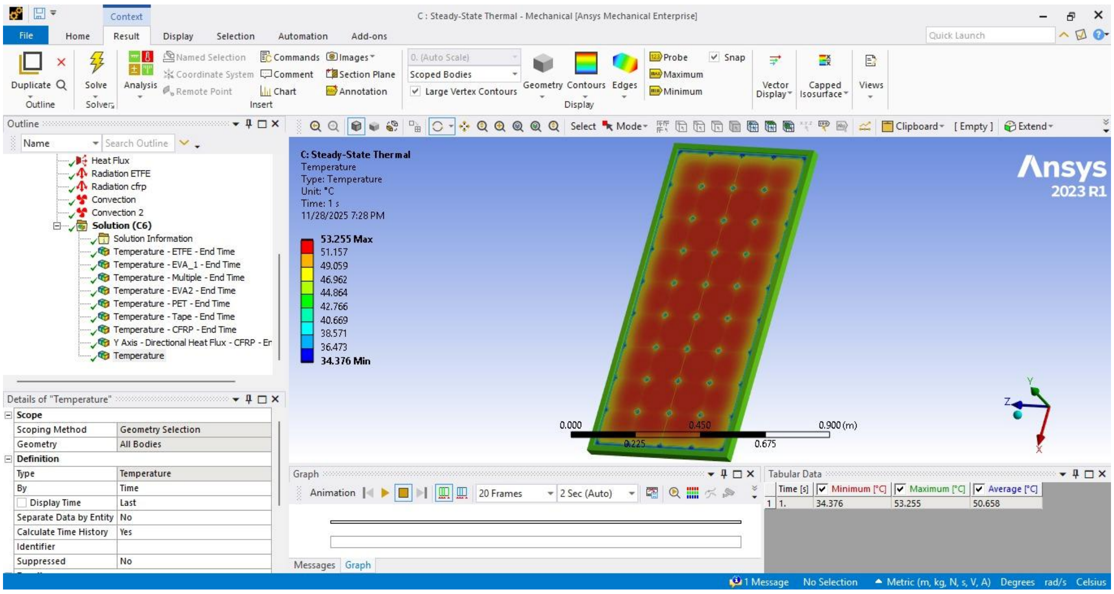


Figure 24: *Temperature distribution contour of the solar PV module*

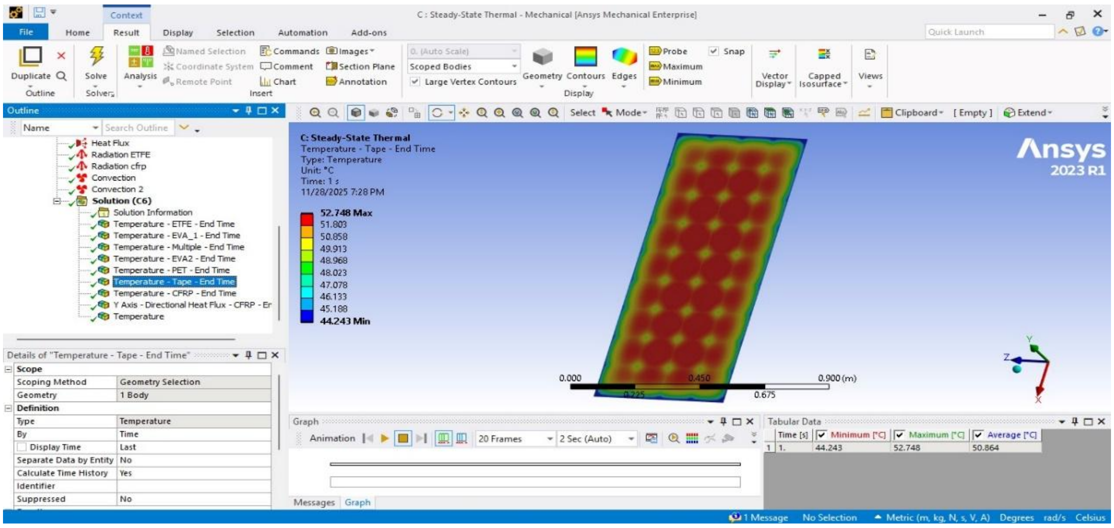


Figure 25: *Temperature distribution contour of the solar PV module TAPE*

14.Discussion

The conduction is found to be the primary mean of internal heat flow, and that heat rejection to the environment is governed by convection and radiation. The inclination angle plays an important role in the convection losses. Appropriate mesh refinement and boundary conditions are essential to obtain accurate thermal predictions. The purpose of the project is to analyze heat transfer characteristics for grid connected centralized solar photovoltaic module. We want to investigate how the temperature distributed in these layers, and evaluate its effect on the system. We developed a detailed model in ANSYS and each layer is described with the actual material properties. In this way our model becomes more reasonable. We were accustomed to normal TA way. The boundary conditions are: temperature, surrounding 25°C (ambient) and emissivity ε =0.89, heat flux fron PV cells of sun. h / 1m2 K=7.95W/m²·K. so This is like real outdoor conditions (according to the references). so This is like actual outdoor conditions (as per the reference). So, our model depicts real-world performance of the PV module.

After simulation, temperature is not uniform. Edges are 44° C, and center values are 52-53° C. Temperature values for PV cells are 46-53° C. ETFE and Eva layers are 34-53° C. Values of CFRP Back Layer are 44-51° C. It concludes that PV cells are more subjected to heat stress than any other component.

Heat flux data indicates the same. Heat flows from the PV cells to the lower levels and thence to the air. This reveals that conduction and convection currents have major roles to remove heat in the module.

When comparing it with other studies and references, it’s quite similar. Refs. give a temperature of 50–55 °C and that PV cells are hotter than other layers. The values of the heat flux are also similar. It’s evident that our mesh (model) is right and performs well.

15.Conclusion

The project examined the thermodynamic characteristics of a solar photovoltaic cell with a focus on the effect of temperature on the efficiency of energy conversion in the cell. The project pointed out the importance of temperature-related factors in the efficiency of silicon solar cells, where a large portion of the solar energy is converted to heat rather than to electrical energy.

The results of the multi-layer laminated structure analysis revealed that temperature distribution in a solar PV module is highly influenced by the properties and thickness of each component layer. As far as heating effects in a solar PV module are concerned, it is clear that a mono-si solar cell is more responsible for solar cell heating, while other layers are responsible for conducting solar cell heating effects. A marginal rise in solar PV module temperature causes a significant decrease in PV efficiency.

The geometry and materials chosen for the panel were also shown to be crucial for thermal performance. Effective encapsulation and support design can go a long way in regulating temperatures and, hence, enhancing both power output and robustness. The results underlined the need to take into consideration thermal effects in designing and optimizing solar panels, especially for applications involving high solar irradiation levels such as in roof-mounted and solar-powered systems.

In summary, this project has shown that the use of thermodynamic analysis is an integral part of understanding and optimizing the performance of photovoltaic cells. The optimization of the thermodynamic conditions of PV cells will enable improved solar energy performance.

16.References

Pavlovic, A., Fragassa, C., Bertoldi, M., & Mikhnych, V. (2021). *Thermal behavior of solar cell encapsulation materials: A numerical and experimental investigation*. Journal of Applied and Computational Mechanics, 7(3), 1847–1855. https://doi.org/10.22055/JACM.2021.37852.3101